BridgeFinder: Finding People Across Structural Holes

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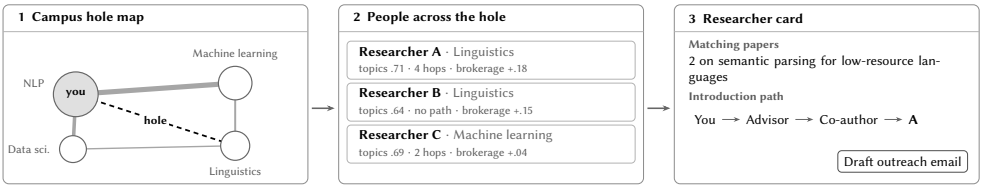


Figure 1: Core flow: (1) co-authorship communities, with a structural hole dashed; (2) ranked researchers across the hole; (3) their matching papers and an introduction path.

1 OBJECTIVE

When students search for a research mentor or researchers seek collaborators, they tend to search through connections they know: their lab, their department, etc. However, novel information spreads through weak ties [5], and individuals spanning structural holes between disconnected communities will be more likely to generate innovative ideas [3]. Interdisciplinary research often harbors these exact structural holes. At UIUC and other universities, separate groups frequently tackle similar problems for years without co-authoring a single paper. The Feb 2024 Gray Book lists 2,574 professor-titled faculty across 110 departments [10], far too many to browse. **BridgeFinder** finds these gaps in the co-authorship graph and names the people on the other side.

2 RELATED WORK

Current tools like Illinois Experts [1] and Google Scholar will match by keyword, but fail to consider graph topology, treating researchers like isolated documents. Barley et al. [2] found that joining an interdisciplinary institute at UIUC diversified members' co-authors, and Dahlander and McFarland [4] studied 15 years of Stanford collaborations, but neither produced a tool. Tang et al. [9] recommend cross-domain collaborators with a topic model but do not use network structure, and link prediction [6] finds likely ties. LinkedIn's "People You May Know" uses triadic closure, which reinforces mutual connections instead of brokerage positioning. BridgeFinder explicitly detects structural holes to recommend collaborators across disconnected communities.

3 COURSE CONCEPTS & NETWORK ANALYSIS

Our project centers on the core concept of **brokerage**, which emphasizes the importance of weak ties that act like local bridges, passing information between two disconnected groups. We will build a graph where the nodes are UIUC researchers in OpenAlex [8], and the edges are co-authorships, weighted by the number of joint papers. In addition, each node will be tagged with the OpenAlex topics that the researcher works on. Researchers who co-author densely with each other form research communities. For a student, we rank researchers at least three steps away by topic similarity, boosted by the number of introduction paths to them. An alternative we considered was triadic closure, but it only finds people already within your network, so it cannot surface new research topics. Our falsifiable claim is that, using only papers from 2015 to 2019, BridgeFinder ranks the actual partners of new cross-community collaborations in 2020 to 2024 higher than triadic closure or topic-only ranking.

4 PROPOSED WORK

We will create a web app with at least the three screens shown in Figure 1; the core flow is described in Section 5. To create the campus map, we will write a Python program to preprocess the UIUC co-authorship graph from OpenAlex (around 110,000 papers from 2015 to 2024).

By Oct 30: We will have the preprocessed graph and communities, version 1 of the app, and feedback from round 1. **By mid-Nov:** We will have the researcher cards, introduction paths, and the round 2 feedback. **By Dec 9:** We will polish the app, and record the final evaluation and video.

Assumptions: the co-authors of papers know each other; OpenAlex correctly identifies people; the OpenAlex topics correctly capture what a researcher worked on; professors who teach a class do research in the corresponding area. The app can't actually tell if a professor is taking students, so it should just be used as a suggestion about people to talk to.

5 PRELIMINARY DESIGN

The core flow of the app follows Figure 1. The student first picks the professors from classes they enjoyed. Then, the map will show their research communities and holes between them. Alongside this, the app will also display a ranked list of researchers across those holes, and each of the researchers can be clicked to open a researcher card that shows their matching papers and the introduction path. Unlike previous work such as Illinois Experts or Google Scholar, which all start with a search box, we collect the user's interests, then show the map first so users can see where the gaps are as well as the overall research landscape. Additionally, we provide reasoning for each suggestion.

6 USERS

Our target user group is UIUC students who liked a class and want to do research in the topic the class covered. A student will be attached to the graph through the professors whose class they enjoyed, and the app will be able to find researchers with similar work (but outside of the professors' circles) along with who could introduce them. We will try to recruit around 10 people (friends and classmates looking for a research position) to test the app through each of the rounds; five to ten testers are enough to surface most usability problems [7], and our larger-scale claims rest on the backtest.

7 EVALUATION

(1) **Backtest:** Looking at the data from 2015 to 2019, we will predict the top 10 relevant people for each researcher using different methods (our method, triadic closure, topic only). We then count how often those people include someone in a different community who they actually started working with in 2020 to 2024.

(2) **Blind rating:** We will give each participant around 15 researcher cards (name, department, recent papers), five from each method. These will be shuffled so they don't know which method produced them. They then rate each researcher based on relevance of their work to their interests, and whether they knew the person already. Our method succeeds if more of its cards are both relevant and new to the participant.

(3) **Usefulness:** We will also survey if our users would feel comfortable enough to contact the suggested people, compared across the three methods.

AI Use: We used Claude as a collaborator to brainstorm ideas, and create an initial draft. Everything was reviewed and verified.

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